



LangChain

By 

LangChain



What is LangChain?

- A framework for building LLM-powered applications.
- Helps combine LLMs with:
 - prompts
 - tools/APIs
 - memory
 - vector databases
 - structured output parsing
 - agents & workflows
- LangChain has 6 core components:
 1. Models
 2. Prompts
 3. Output Parsers
 4. Memory
 5. Tools
 6. Agents



Components

1. Models

- Chat models
- Text completion models
- Embedding models
- Multi-modal models

Can Use ChatOpenAI / ChatGroq / ChatAnthropic

2. Prompts

- Prompt Templates
- ChatPromptTemplate
- Few-shot prompting
- Prompt variables

Components



3. Output Parsers

- Why structured output? (JSON, lists, tables)
- PydanticOutputParser
- StructuredOutputParser

Ex: Force LLM to return exact JSON schema.

4. Memory

Short-term conversation memory

Types:

- Buffer memory
- Summary memory
- Entity memory
- Vector store memory

Components



5. Tools

- Tools = functions/models/APIs LLM can call
- Built-in tools (calculator, search APIs)
- Custom tools

6. Agents

- Agent = LLM + Tools + Reasoning
- ReAct reasoning
- Types:
 - Structured agents
 - ToolCalling agents
 - OpenAI-style Agents

LangChain's SDK



LangChain has decomposed into smaller packages.

1. Langchain-core:

This is the **foundation layer**. This is the execution backbone.

- Base abstractions
- Runnables
- Prompt templates
- Output parsers
- Message objects
- Tool definitions

2. Langchain :

This is the higher-level application layer. This is application logic.

- Chains
- Agents
- Memory implementations
- RAG utilities
- Classic abstractions

This builds on top of langchain-core.

3. Provider-Specific packages:

LangChain separated integrations into dedicated packages:

Examples:

- langchain-openai
- langchain-google-genai
- langchain-anthropic
- langchain-mistralai

These wrap specific providers like:

- OpenAI
- Google
- Anthropic

Why separate?

- Independent versioning
- Faster updates
- Reduced dependency conflicts

4. Langchain-community:

Earlier vectorDB implementation and third party api provider modules are in this package but later they are separated into individual provider specific packages. Still what it contains

- Smaller integrations
- Experimental connectors
- Less actively maintained tools
- Integrations without dedicated packages

A large, abstract graphic consisting of several concentric, overlapping rings in shades of blue, green, and purple, framing the central text.

Thank you